Dear Dr. X,

Please consider our paper, entitled "Biological and climatic constraints drive reproductive synchrony" for publication as a X in *Nature Plants*.

Understanding how plants time reproduction under a fluctuating and unpredictable climate is a fundamental question in plant biology. This timing shapes not only individual reproductive success, but also the reproductive dynamics of whole populations. Across temperate and tropical forests, many species have individual trees that align their cycles and reproduce together with seed crops that vary a lot from year to year—a synchronous phenomenon called masting.

What sustains this synchrony depends on how climate interacts with the reproductive biology of individual trees. Reproduction is limited by physiological constraints and resources that prevent trees from reproducing freely every favorable year. How these individual-level constraints scale up to shape reproduction across populations remains poorly understood. This gap limits our understanding of how reproduction is regulated within trees, and reduce our ability to forecast how forests will regenerate as the climate warms.

Building on decades of research on tree reproductive biology and long-term seed production data, we develop an approach that links individual-tree reproduction to population-scale dynamics. In particular, we integrate a well-established reproductive constraint, the temporal overlap between fruit maturation and floral bud initiation, with climatic drivers of masting. [...]

We show that reproductive constraints buffer against the runaway effects of climate change. Even under major warming, trees cannot remain locked in a high reproductive state, preventing major shifts in the frequency of high seed-production years. At the population scale, the interplay between individual constraints and shared climatic cues provides a simple physiological mechanism for synchrony. Cold summers act as the critical hinge, driving most trees into a low reproductive state, and [setting the stage] for a synchronous mast year when a warm summer follows.

Our results advance fundamental plant biology by showing how individual-level constraints and population-level climatic cues interact to produce synchrony that can persist over multiple years. Contrary to the expectation that warming is already disrupting masting, we find that synchrony has remained stable over four decades of significant warming, suggesting that it may be more resilient than previously predicted. However, synchrony might decline with further warming, because the cold summers that synchronize trees could become increasingly rare.

Our work illustrates a principle that extends beyond forests and masting. Plant biological constraints can fundamentally limit the runaway amplification of climate effects on emergent phenomena at larger scales. Developing a mechanistic understanding of the biological and climatological drivers of plant timing will be critical to forecast regeneration and to understand the long-term adaptive responses of plant communities more broadly.

All authors contributed to this work and approve this version for submission. The manuscript is X words, with X references and X figures, and is not under consideration elsewhere. We hope you find it suitable for publication in *Nature Plants*, and look forward to hearing from you.

Sincerely,

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References